

Frequency Calibration of HackRF using the GSM FCCH “Tone” (Mathematical process in 2 pages)

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February 10, 2026

1. Physical Concept and Modeling

In an SDR like the HackRF, an error in the local oscillator (LO) manifests as a **frequency offset** that shifts the *entire* received signal. If we tune a carrier to frequency f_c but the LO is offset, we will observe an additional translation Δf in baseband.

Ideal Reception Model (without error). Let $r_{\text{RF}}(t)$ be the RF signal centered at f_c and let the ideal mixer multiply by $e^{-j2\pi f_c t}$. In ideal baseband:

$$x(t) = r_{\text{RF}}(t) e^{-j2\pi f_c t}. \quad (1)$$

Model with LO Error. If the actual LO is $f_c + \Delta f$ (due to oscillator error), the mixing process is:

$$x_{\Delta}(t) = r_{\text{RF}}(t) e^{-j2\pi(f_c + \Delta f)t} = x(t) e^{-j2\pi \Delta f t}. \quad (2)$$

Therefore, **the LO error appears as a complex rotation** $e^{-j2\pi \Delta f t}$ in baseband, equivalent to a spectral shift of Δf .

2. Why GSM is Suitable: The FCCH Burst as a Tone

In GSM, there is a **Frequency Correction Channel (FCCH)** that periodically transmits the **Frequency Correction Burst (FB)**. Its sequence is designed so that, after GMSK modulation, the baseband result is a **nearly pure tone** (a very narrow spectral peak) at a known frequency:

$$f_{\text{FCCH}} = \frac{R_s}{4}, \quad (3)$$

where the GSM symbol rate is

$$R_s = \frac{1.625 \times 10^6}{6} \approx 270,833.33 \text{ sym/s}. \quad (4)$$

Thus,

$$f_{\text{FCCH}} \approx \frac{270,833.33}{4} \approx 67,708.33 \text{ Hz}. \quad (5)$$

In summary: if you tune to a GSM channel and capture IQ data, you should see a peak near 67.708 kHz in baseband during an FCCH.

3. Mathematical Process for Estimating the Offset Δf

The objective is to estimate Δf by measuring the shift of the FCCH tone relative to f_{FCCH} .

3.1. Capture and Temporal Selection

Capture a window of IQ (baseband) samples $x_{\Delta}[n]$ sampled at frequency F_s :

$$x_{\Delta}[n] = x_{\Delta}(t) \Big|_{t=n/F_s}, \quad n = 0, \dots, N-1. \quad (6)$$

Ideally, select a segment where the FCCH is present (otherwise, you can search for the most dominant peak near f_{FCCH}).

3.2. Spectral Estimation (FFT/PSD)

Calculate the DFT:

$$X_{\Delta}[k] = \sum_{n=0}^{N-1} x_{\Delta}[n] e^{-j2\pi kn/N}, \quad k = 0, \dots, N-1. \quad (7)$$

Map each bin to a frequency (centered at 0) using the FFT shift re-indexing:

$$f[k] = \left(k - \frac{N}{2}\right) \frac{F_s}{N}, \quad k = 0, \dots, N-1. \quad (8)$$

Then estimate the spectral magnitude $|X_{\Delta}[k]|$ (or PSD). The FCCH appears as a peak:

$$k^* = \arg \max_{k \in \mathcal{K}} |X_{\Delta}[k]|, \quad (9)$$

where $\mathcal{K}$ is a neighborhood around f_{FCCH} (for example, ± 20 kHz) to avoid confusing it with interference.

Measured Tone Frequency.

$$\hat{f}_{\text{tono}} = f[k^*]. \quad (10)$$

3.3. Calculation of the Absolute Offset in Hz

If there were no LO error, the peak should be at f_{FCCH} (eq. (5)). With error, per (2), the peak shifts approximately by Δf :

$$\hat{f}_{\text{tono}} \approx f_{\text{FCCH}} + \Delta f. \quad (11)$$

Therefore, the estimator for the offset in Hz is:

$$\widehat{\Delta f} = \hat{f}_{\text{tono}} - f_{\text{FCCH}}. \quad (12)$$

3.4. Conversion to ppm (Parts Per Million)

The standard oscillator error specification is expressed in ppm:

$$\widehat{\text{ppm}} = \frac{\widehat{\Delta f}}{f_c} \times 10^6, \quad (13)$$

where f_c is the central RF frequency of the GSM channel (the carrier you tuned to).

Numerical Example. If you tune to $f_c = 947.6$ MHz and measure $\widehat{\Delta f} = 5$ kHz:

$$\widehat{\text{ppm}} = \frac{5000}{947.6 \times 10^6} \times 10^6 \approx 5.28 \text{ ppm}.$$

4. How to Apply the Correction

There are two equivalent ways:

4.1. Correcting the Tuning Frequency

If your SDR does not have a ppm correction parameter, you can compensate by adjusting the tuning:

$$f_{\text{tune}} = f_{\text{objetivo}} - \widehat{\Delta f} \iff f_{\text{tune}} \approx f_{\text{objetivo}} \left(1 - \frac{\widehat{\text{ppm}}}{10^6}\right). \quad (14)$$

4.2. Correcting in Baseband (Digital Mixing)

Multiply the signal by a complex exponential that cancels the rotation:

$$y[n] = x_{\Delta}[n] e^{+j2\pi \widehat{\Delta f} n / F_s}. \quad (15)$$

This shifts the spectrum back to its place and “straightens” the tone.

5. Practical Notes (for best results)

- **Spectral resolution:** $\Delta f_{\text{bin}} = F_s/N$. To better estimate $\widehat{\Delta f}$, use a large N (or average several FFTs).
- **Thermal drift:** Δf can change with temperature. It is advisable to calibrate after a few minutes of warm-up.
- **Band dependency:** the offset may vary slightly with the band/PLL. Ideally, calibrate near your band of interest (not just on any random GSM channel).
- **Best solution: external clock:** using a 10 MHz reference (GPSDO/OCXO) at CLKIN improves both offset and stability.

6. Summary of the Mathematical Pipeline

1. Capture $x_{\Delta}[n]$ (IQ) at F_s on a GSM channel with FCCH.
2. Estimate the spectrum $|X_{\Delta}[k]|$ and locate the peak: $\hat{f}_{\text{tono}}$.
3. Calculate the offset in Hz: $\widehat{\Delta f} = \hat{f}_{\text{tono}} - f_{\text{FCCH}}$.
4. Convert to ppm: $\widehat{\text{ppm}} = \widehat{\Delta f}/f_c \cdot 10^6$.
5. Correct the tuning or perform digital mixing to compensate for $\widehat{\Delta f}$.